

DLD HOMEWORK #2

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10303

Q1:-

$$a) F = (a+b+c+d)(a+b'+c+d)(a'+b'+c'+d')$$

f complement will be-

$$F' = \overline{(a+b+c+d)(a+b'+c+d)(a'+b'+c'+d')}$$

$$\because \overline{AB} = \overline{A} + \overline{B}$$

$$F' = \overline{(a+b+c+d)} + \overline{(a+b'+c+d)} + \overline{(a'+b'+c'+d')}$$

$$F' = \overline{a} \overline{b} \overline{c} \overline{d} + \overline{a} \overline{b} \overline{c} d + \overline{a} \overline{b} c \overline{d} + \overline{a} \overline{b} c d + \overline{a} b \overline{c} \overline{d} + \overline{a} b \overline{c} d + \overline{a} b c \overline{d} + \overline{a} b c d + a \overline{b} \overline{c} \overline{d} + a \overline{b} \overline{c} d + a \overline{b} c \overline{d} + a \overline{b} c d + a b \overline{c} \overline{d} + a b \overline{c} d + a b c \overline{d} + a b c d$$

$$F' = \overline{a} \overline{c} \overline{d} (\overline{b} + b) + a b c d$$

$$F' = \overline{a} \overline{c} \overline{d} + a b c d$$

b)

$$i) (A'+B)(A+B')(AC)$$

apply distributive law- $(A'+B)(A+B')$

$$(A'A + A'B' + BA + BB') (AC)$$

$$(A'B' + AB) (AC)$$

$$(A \cdot A') B' \cdot C + A \cdot B \cdot C$$

A.B.C $\rightarrow$ Has three literals.

$$\text{ii) } (z)(w+x')(wx'yz + wz)$$

Using null law:-

$$(z)(w+x')(wz)(\cancel{x'y+1}) \rightarrow 1$$

$$\therefore z.z = z$$

$$(wz)(w+x')$$

$$\cancel{w} \cdot \overset{w}{w} \cdot z + x' \cdot w \cdot z$$

$$wz + wzx'$$

$$wz(\cancel{1+x'}) \rightarrow 1$$

$wz \rightarrow$ two literals.

$$\text{iii) } (xy+z') + (xz'y+y)'$$

$$xy+z' + \overline{xz'y} \cdot \overline{y}$$

$$xy+z' + (\overline{x+z+y'}) \overline{y}$$

$$xy+z' + \overline{x} \overline{y} + z \overline{y} + \overline{y} \cdot \overline{y}$$

$$xy + \overline{z} + \overline{x} \overline{y} + z \overline{y} + \overline{y}$$

$$xy + \overline{z} + \overline{y}(\cancel{\overline{x} + z + 1}) \rightarrow 1$$

$$xy + \overline{z} + \overline{y} \rightarrow 4 \text{ literals}$$

$$\cancel{\overline{x} + \overline{y} + \overline{z} + \overline{y}} \rightarrow \overline{x} + \overline{y} + \overline{z} \quad \text{or} \quad x y z$$

Three literals



c) $A = 10001110$
 $B = 11001101$

- A and B:- returns one if Both sides are one, otherwise zero.

$$\begin{array}{r} A = 10001110 \\ B = 11001101 \end{array}$$

$$A \text{ and } B = 10001100$$

- A XOR B:- returns one if Both bits are different.

$$\begin{array}{r} A = 10001110 \\ B = 11001101 \end{array}$$

$$A \text{ XOR } B = 01100001$$

- A XNOR B:- returns one if Bits are same, else zero-

$$\begin{array}{r} A = 10001110 \\ B = 11001101 \end{array}$$

$$A \text{ XNOR } B = 10111000$$

Q21-

$$i) xy + x(y' + z) + xyz'$$

using distributive law:- $x(y' + z)$:-

$$xy + xy' + xz + xyz'$$

taking common-

$$x(y + y') + xz + xyz'$$

using inverse law $(y + y') = 1$

$$x \cdot 1 + xz + xyz'$$

$$x + xz + xyz'$$

taking common-

$$x(1 + z) + xyz'$$

using Null law $(1 + z) = 1$

$$x + xyz'$$

taking common-

$$x(1 + yz')$$

Null law:-

$$\boxed{x} \text{ Ans}$$

$$ii) w(x+y)(x+y')(w'+x'+z)$$

$$(wx + wy)(x+y')(w'+x'+z)$$

applying distributive law-

$$(wx \cdot x + wx \cdot y + wy \cdot x + wy \cdot y)(w' + x' + z)$$

Applying inverse law $\{y + y' = 1\}$ idempotent-

$$(wx + wx \cdot y + wx \cdot y)(w' + x' + z)$$

$$(wx + wx(y + y'))(w' + x' + z)$$

applying inverse law-

$$(wx + wx)(w' + x' + z)$$

applying idempotent law-

$$(wx)(w' + x' + z)$$

applying distributive law-

$$(w \cdot w' \cdot x + w \cdot x \cdot x' + wxz)$$

apply inverse law-

$$\boxed{wxz} \text{ Ans}$$

$$\text{iii) } (xy + x')' + xyz' + z$$

apply de Morgan's law:-

$$(\overline{xy})x + xy\overline{z} + z$$

apply de Morgan's law:-

$$(\overline{x} + \overline{y})x + xy\overline{z} + z$$

$$\overline{x}x + x\overline{y} + xy\overline{z} + z$$

inverse law:-

$$x\overline{y} + xy\overline{z} + z$$

$$x(\overline{y} + y\overline{z}) + z$$

apply distributive law:-

$$A + BC = (A+B) \cdot (A+C)$$

$$x[(\overline{y} + y) \cdot (\overline{y} + \overline{z})] + z$$

$$x(\overline{y} + \overline{z}) + z$$

$$x\overline{y} + x\overline{z} + z \rightarrow 5 \text{ literals}$$

Q3:-

3 Different keys $\rightarrow a, b, c$ Door $\rightarrow z$ Alarm $\rightarrow k$

a) Truth Table-

a	b	c	z	k	Terms
0	0	0	0	0	m_0
0	0	1	0	1	m_1
0	1	0	0	1	m_2
0	1	1	1	0	m_3
1	0	0	0	1	m_4
1	0	1	1	1	m_5
1	1	0	1	0	m_6
1	1	1	0	1	m_7

b) For minimum value of k - (SOP)

$$f(k)_{\min} = m_1 + m_2 + m_4 + m_5 + m_7$$

$$= \bar{a}\bar{b}c + \bar{a}b\bar{c} + a\bar{b}\bar{c} + a\bar{b}c + abc$$

$$= \bar{a} + b + \bar{c} + a + \bar{b} + c$$

For maximum function of k - (POS)

$$f(k)_{\max} = (a + b + c) \cdot (a + b' + c') \cdot (a' + b' + c)$$

Q41-

$$i) F = (x+z') + (x+y') (x+y+z')$$

$$\bar{F} = \overline{(x+z') + (x+y') (x+y+z')}$$

$$\bar{F} = \overline{(x+z')} \cdot \overline{(x+y') (x+y+z')}$$

$$\bar{F} = \overline{xz} \left[\overline{(x+y') + (x+y+z')} \right]$$

$$\bar{F} = \bar{x}z \left[\bar{x}y + \bar{x}\bar{y}z \right]$$

$$\bar{F} = \bar{x} \cdot \bar{x} \cdot y \cdot z + \bar{x} \cdot \bar{x} \cdot \bar{y} \cdot z \cdot z$$

$$\bar{F} = \bar{x}yz + \bar{x}\bar{y}z$$

$$\bar{F} = \bar{x}z (y + \bar{y})$$

$$\boxed{F = \bar{x}z}$$

$$ii) F = (xyz' + x'z + y+z') (x+y)$$

$$\bar{F} = \overline{(xyz' + x'z + y+z') (x+y)}$$

$$\bar{F} = \overline{(xyz' + x'z + y+z')} + \overline{(x+y)}$$

$$\bar{F} = (\bar{x}yz + \bar{x}z + y + \bar{z}) + (\bar{x} + \bar{y})$$

$$\bar{F} = (\bar{x} + \bar{y} + \bar{z}) (\bar{x} + \bar{z}) (\bar{z} + \bar{y}) + \bar{x} \bar{y}$$

$$\bar{F} = (\bar{x}yz + \bar{y}z + \bar{x}z) (\bar{x} + \bar{y} + \bar{z}) + \bar{x} \bar{y}$$

$$\bar{F} = (x \bar{y} z) (\bar{x} + \bar{y} + \bar{z}) + x \bar{y}$$

$$\bar{F} = (\cancel{x \bar{x} \bar{y} z} + x \bar{y} z + \cancel{x \bar{y} z \bar{z}}) + (x \bar{y})$$

$$\bar{F} = x \bar{y} z + x \bar{y}$$

$$\boxed{F = \bar{y} (x z + \bar{x})} \quad \text{Ans}$$

Q51-

$$F(A, B, C, D) = (\bar{A} \bar{B} \bar{D}) + (\bar{A} \bar{C}) + (\bar{B} \bar{D}) + \bar{A} C$$

a) The following equation is sum of Product (SOP) form

b)

$$\bar{A} \bar{B} \bar{D} + \bar{C} \bar{D} + \bar{A} (\bar{C} + C)$$

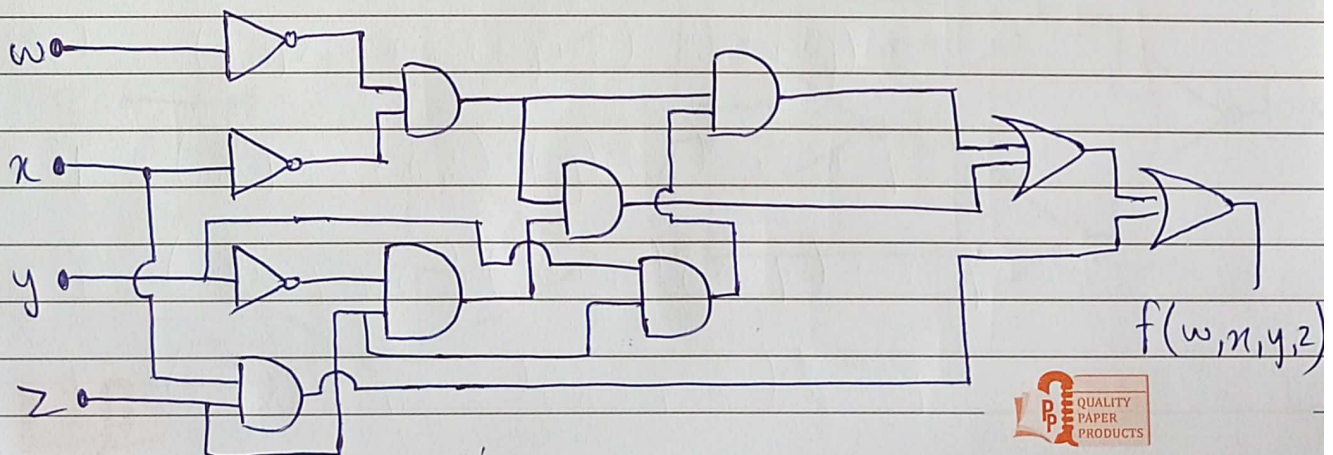
$$\bar{A} (\bar{B} \bar{D} + 1) + \bar{C} \bar{D}$$

$$F(A, B, C, D) = \boxed{\bar{A} + \bar{C} \bar{D}} \quad \text{Ans} \rightarrow 3 \text{ literals}$$

Q6r $F(w, x, y, z) = w'x'y'z + xz + w'x'y z$

w	x	y	z	$f(w, x, y, z)$
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	1	0
1	0	1	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

b) logic diagram:-



c) $f(w, x, y, z) = w'x'y'z + xz + w'x'yz$

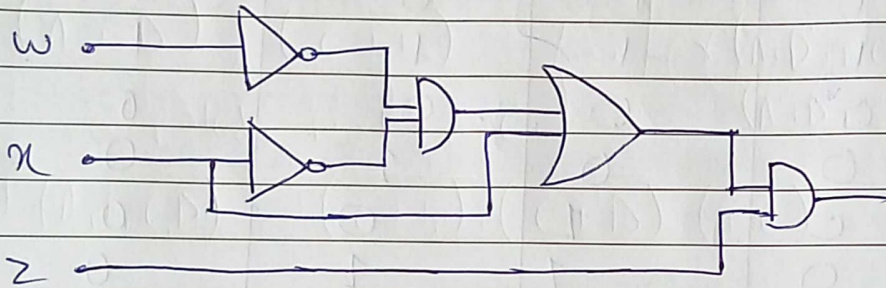
factoring:-

$$= w'x'z(y' + y) + xz$$

$$= w'x'z + xz$$

$$f(w, x, y, z) = z(w'x' + x)$$

d)



gates used in this one are 5, but gates used in part B are 11, hence less than half gates were used in this one.

Q81-

$$a) F(A, B, C, D) = (AD + C'B')' \quad [POS]$$

$$F = \overline{(AD + C'B')}$$

using De Morgan's law:-

$$F = \overline{AD} \cdot \overline{C'B'}$$

$$F = (\overline{A} + \overline{D}) \cdot (C + B) \quad \text{Ans}$$

completed next page.

$$\rightarrow (\overline{A} + \overline{D} + B + C)(\overline{A} + \overline{D} + B' + C') \cdot (\overline{A} + \overline{D} + B + C)(\overline{A} + \overline{D} + B' + C')$$

• Completes part 8(a)

adding B, C in $(\bar{A} + \bar{D})$:-

$$= (\bar{A} + \bar{D} + B + C) (\bar{A} + \bar{D} + \bar{B} + C) (\bar{A} + \bar{D} + B + \bar{C}) (\bar{A} + \bar{D} + \bar{B} + \bar{C})$$

adding A, D in $(C + B)$:-

$$= (A + D + C + B) (\bar{A} + \bar{D} + C + B) (A + \bar{D} + C + B) (\bar{A} + \bar{D} + C + B)$$

$$\text{all terms} = (A + B + \bar{C} + D) (A + B + C' + D') (A + B' + C + D) (A + \bar{B} + C + \bar{D}) \\ (A + \bar{B} + C + D) (A + \bar{B} + C' + D') (A' + B + C' + D) (A' + B' + C + D) \\ (A' + B' + C' + D)$$

$$b) F(w, x, y, z) = (y'z + x)(xy' + wy')(z' + yw) \quad [\text{SOP}]$$

$$F = (\bar{y}z + x)(x\bar{y} + w\bar{y})(\bar{z} + yw)$$

$$F = (\bar{y}z + x)\bar{y}(x + w)(\bar{z} + yw)$$

$$F = (\bar{y}z + x)(x + w)(\bar{y}\bar{z})$$

$$F = \cancel{\bar{y}} x \bar{z} \bar{y} (x + w)$$

$$\boxed{\cancel{F = x\bar{y}z + wx\bar{y}z}} \quad \text{Ans. } F = x\bar{z}\bar{y} \quad (\text{Absorption law.})$$

addig:- $\boxed{x\bar{y}\bar{z}w + w'x\bar{y}'z'}$

$$d) F'(w, x, y, z) = w'x'y + x'y'z + w'y'z$$

$$c) F(A, B, C, D) = \overline{(C+B)} \overline{(C+D)} (BC) \quad [SOP]$$

Using De Morgan's law:-

$$F = \overline{(C+B)} \overline{(C+D)} (BC)$$

$$F = (\overline{C} \overline{B}) (\overline{C} \overline{D}) (BC)$$

$$F = (\overline{C} \cancel{C}) (\overline{B} \cancel{B}) (\overline{D})$$

$$\boxed{F = 0} \quad \text{No terms}$$



$$d) F'(\omega, \kappa, y, z) = \omega' \kappa' y + \kappa' y' z + \omega' y' z$$

To make F into POS form, we need complement of F :-

$$F'' = (\bar{\omega} \bar{\kappa} y + \bar{\kappa} \bar{y} z + \bar{\omega} \bar{y} z)$$

$$F = (\bar{\omega} \bar{\kappa} y) (\bar{\kappa} \bar{y} z) (\bar{\omega} \bar{y} z)$$

$$F = (\omega + \kappa + \bar{y}) (\kappa + y + \bar{z}) (\omega + y + \bar{z}) \text{ Ans}$$



e)

$$F(A, B, C, D) = (B' + C' + D')(A' + B')(A' + B')$$

Let $A' + B' = K$

$$F = (B' + C' + D')(K)(K)$$

By idempotent law we know $K \cdot K = K$

$$F = (B' + C' + D')(K)$$

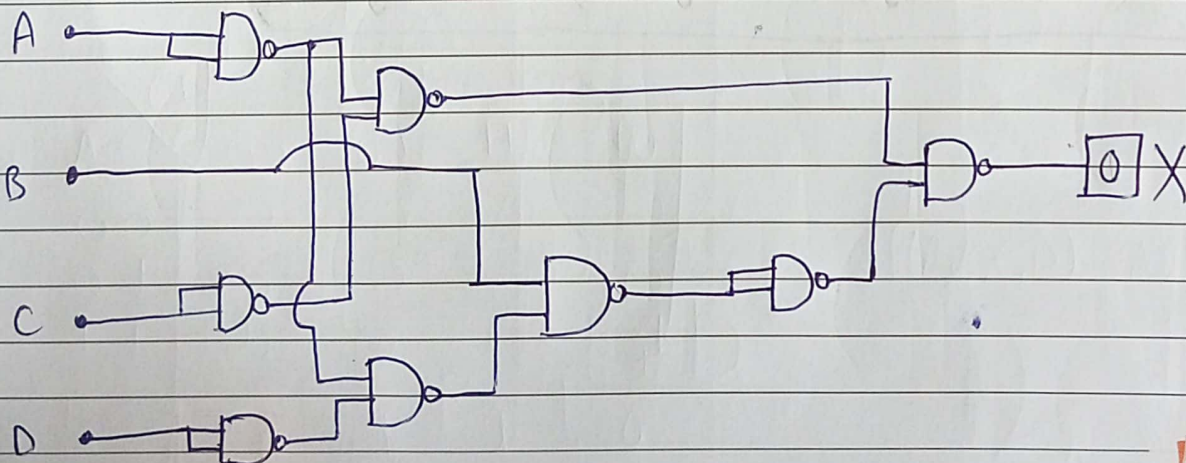
$$F = (B' + C' + D')(A' + B')$$

~~$$F = A'B' + A'C' + A'D' + B' + B'C' + B'D'$$~~

~~$$F = A'B' + A'C' + A'D' + B' + B'C' + B'D'$$~~

Using absorber law- $A + AB = A$

$$F = C'A' + D'A' + B'$$



Q91-

$$b) F(A, B, C, D) = \sum (1, 2, 3, 6, 13, 15)$$

A	B	C	D	F	
0	0	0	0	0	m_0
0	0	0	1	1	m_1
0	0	1	0	1	m_2
0	0	1	1	1	m_3
0	1	0	0	0	m_4
0	1	0	1	0	m_5
0	1	1	0	1	m_6
0	1	1	1	0	m_7
1	0	0	0	0	m_8
1	0	0	1	0	m_9
1	0	1	0	0	m_{10}
1	0	1	1	0	m_{11}
1	1	0	0	0	m_{12}
1	1	0	1	1	m_{13}
1	1	1	0	0	m_{14}
1	1	1	1	1	m_{15}

AB \ CD	00	01	11	10
00	0	1	1	1
01	0	0	0	1
11	0	1	1	0
10	0	0	0	0

$$f(A, B, C, D) = A'B'D + \cancel{A'B'D} + CD'A' + ABD$$

$$F(A, B, C, D) = A'B'D + \cancel{A'B'C} + CDA + ABD$$

2)

$$c) F(A, B, C, D) = \sum(9, 11, 14)$$

A	B	C	D	F	
0	0	0	0	0	m_0
0	0	0	1	0	m_1
0	0	1	0	0	m_2
0	0	1	1	0	m_3
0	1	0	0	0	m_4
0	1	0	1	0	m_5
0	1	1	0	0	m_6
0	1	1	1	0	m_7
1	0	0	0	0	m_8
1	0	0	1	1	m_9
1	0	1	0	0	m_{10}
1	0	1	1	1	m_{11}
1	1	0	0	0	m_{12}
1	1	0	1	0	m_{13}
1	1	1	0	1	m_{14}
1	1	1	1	0	m_{15}

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	0	0	1
10	0	1	1	0

$$f(A, B, C, D) = AB'D + ABCD'$$

$$a) F(w, x, y, z) = \pi(2, 3, 7, 8, 13, 14)$$

w	x	y	z	F	
0	0	0	0	1	m_0
0	0	0	1	1	m_1
0	0	1	0	0	m_2
0	0	1	1	0	m_3
0	1	0	0	1	m_4
0	1	0	1	1	m_5
0	1	1	0	1	m_6
0	1	1	1	0	m_7
1	0	0	0	0	m_8
1	0	0	1	1	m_9
1	0	1	0	1	m_{10}
1	0	1	1	1	m_{11}
1	1	0	0	1	m_{12}
1	1	0	1	0	m_{13}
1	1	1	0	0	m_{14}
1	1	1	1	1	m_{15}

~~| w \ x \ y \ z | 00 | 01 | 11 | 10 |
|---------------|----|----|----|----|
| 00 | 1 | 1 | 0 | 0 |
| 01 | 1 | 1 | 0 | 1 |
| 11 | 1 | 0 | 1 | 0 |
| 10 | 0 | 1 | 1 | 1 |~~

~~$$f(w, x, y, z) = \sum m(2, 3, 7, 8, 13, 14)$$~~

~~$$f(w, x, y, z) = (w + x + y + z)$$~~

w \ x \ y \ z	00	01	11	10
00	1	1	0	0
01	1	1	0	1
11	1	0	1	0
10	0	1	1	1

$$f(w, x, y, z) = (w' + x + y + z)(w' + x' + y + z')(w + y' + z')(w + x + y') \left(\frac{w' + x' + y' + z}{(w' + x' + y' + z)} \right)$$

$$f(w, x, y, z) = (w' + x + y + z)(w' + x' + y + z')(w + y' + z')(w + x + y')(w' + x' + y' + z)$$

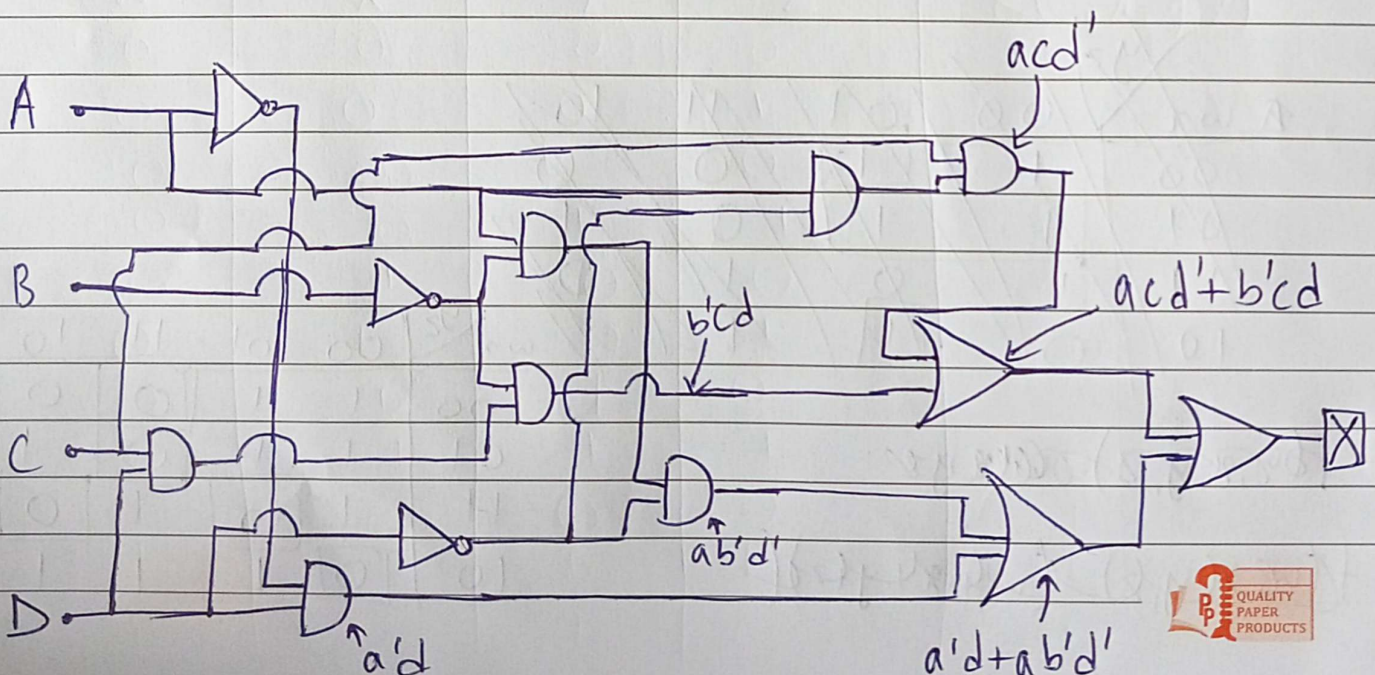
Answer.

Q101- $X(A, B, C, D) = \Sigma(1, 2, 3, 5, 7, 8, 10, 14)$

K map:-

ab \ cd		cd			
		00	01	11	10
ab	00	0	1	1	1
	01	0	1	1	0
	11	0	0	0	1
	10	1	0	0	1

$$X(A, B, C, D) = a'd + ab'd' + b'cd + acd'$$



$$Y(A, B, C, D) = \pi(0, 1, 3, 15, 11, 12)$$

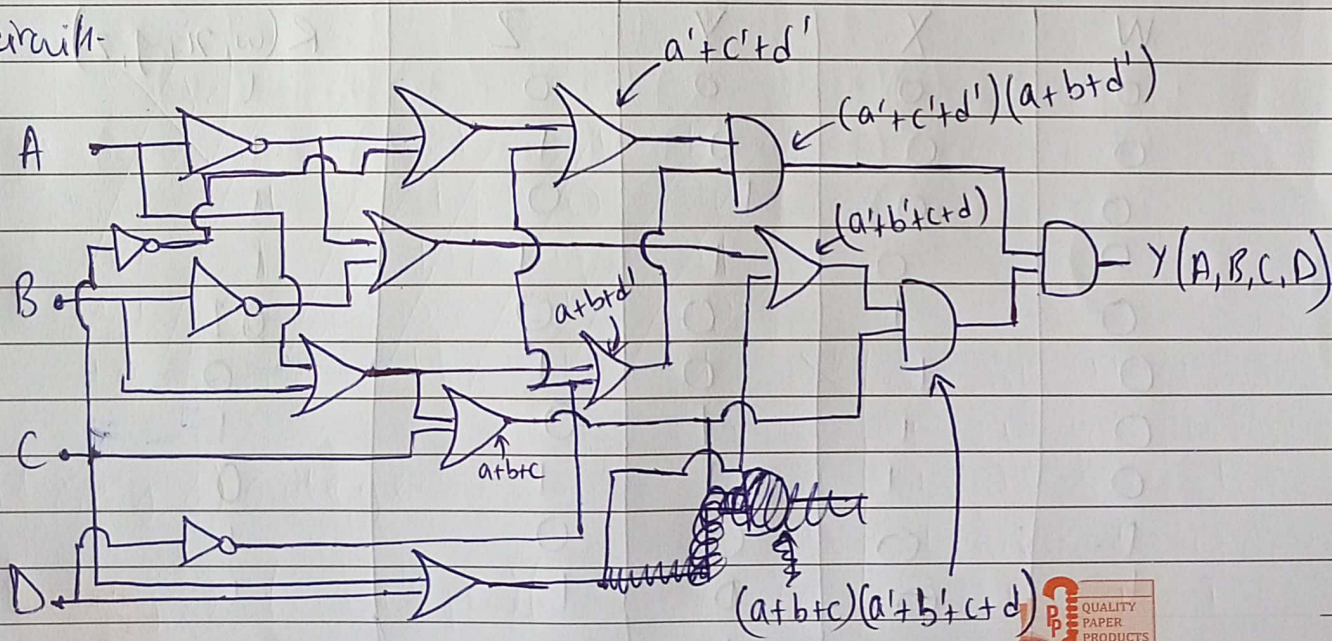
AB \ CD	00	01	11	10
00	1	0	0	1
01	1	1	1	1
11	0	1	0	1
10	1	1	0	1

$$Y(A, B, C, D) = (a' + b' + c + d) \cdot (a + b + d) \cdot (a + b + d') \cdot (a' + c' + d')$$

~~$$Y = (a'd' + ac + ad' + a'b + bc' + bd' + a'c + c'd' + cd')$$~~

~~$$Y = (a'c' + ac + ad' + a'b + bc' + bd' + cd')$$~~

circuit:



Q11- according to the circuit-

a) the equation:-

$$r(w, x, y, z) = \overline{(w' + x'y')} \cdot \overline{(x' + x'y')} \cdot (z + x'y')$$

using De Morgan's law:-

$$r = \overline{(w' + x'y')} + \overline{(x' + x'y')} + (z + x'y')$$

$$r = (w \cdot \overline{x} \cdot \overline{y}) + \overline{x} \cdot (1 + \overline{y}) + (\overline{z} \cdot \overline{x} \cdot \overline{y})$$

$$r = w(x+y) + \overline{x} + \overline{z}(x+y)$$

$$r = w(x+y) + \overline{x} + \overline{z}(x+y)$$

$$\boxed{r = (x+y)(w+\overline{z}) + \overline{x}} \text{ Ans.}$$

b)

W	X	Y	Z	$R(w, x, y, z)$
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

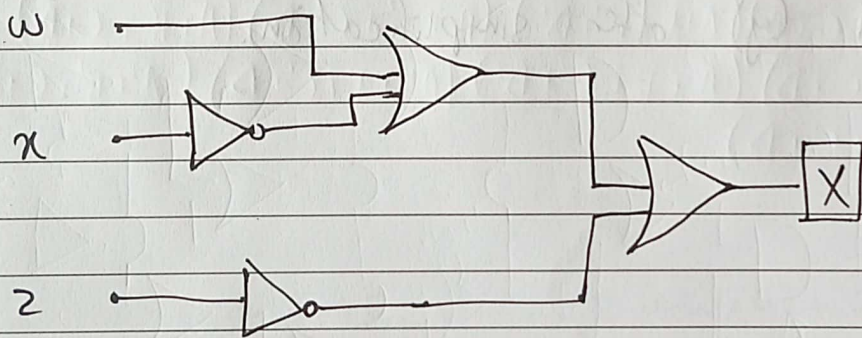
w	x	y	z	$r(w, x, y, z)$
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

c)

wx \ yz	00	01	11	10
00	1	1	1	1
01	1	0	0	1
11	1	1	1	1
10	1	1	1	1

equation:- $w + \bar{x} + \bar{z}$

circuit:-



Ques d:-

w	x	y	z	$w+x'+z'$
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

No, y input doesn't affect the output of circuit, we don't need (y) after simplification.

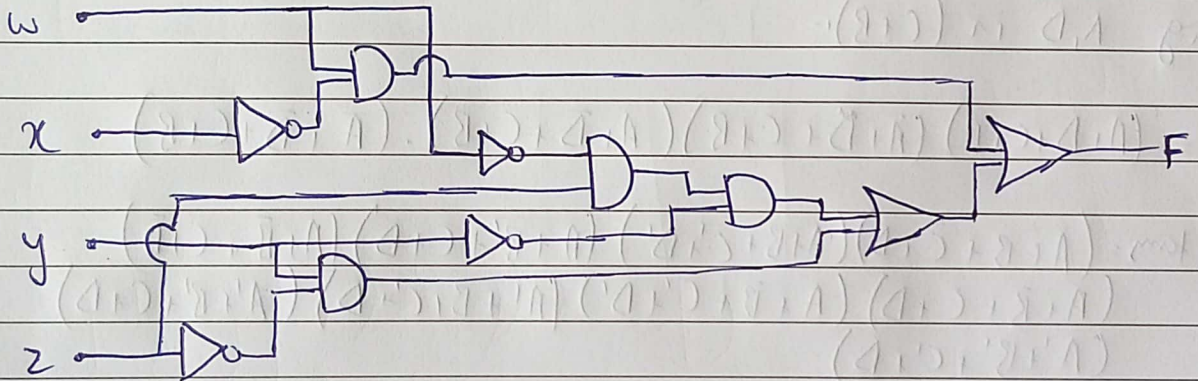
Q12

a) $F = \sum (1, 2, 5, 8, 10, 14)$
 $d(3, 6, 9, 11)$

$wx \backslash yz$	00	01	11	10
00	0	1	X	1
01	0	1	0	X
11	0	0	0	1
10	1	X	X	1

$$F(w, x, y, z) = w\bar{x} + y\bar{z} + \bar{w}\bar{y}z$$

b)



c)

$$F = \bar{w}\bar{y}\bar{x}\bar{z} + \bar{w}x\bar{y}z + \bar{w}x\bar{y}z + w\bar{x}\bar{y}\bar{z} + \bar{w}x\bar{y}\bar{z} + w\bar{x}\bar{y}\bar{z}$$

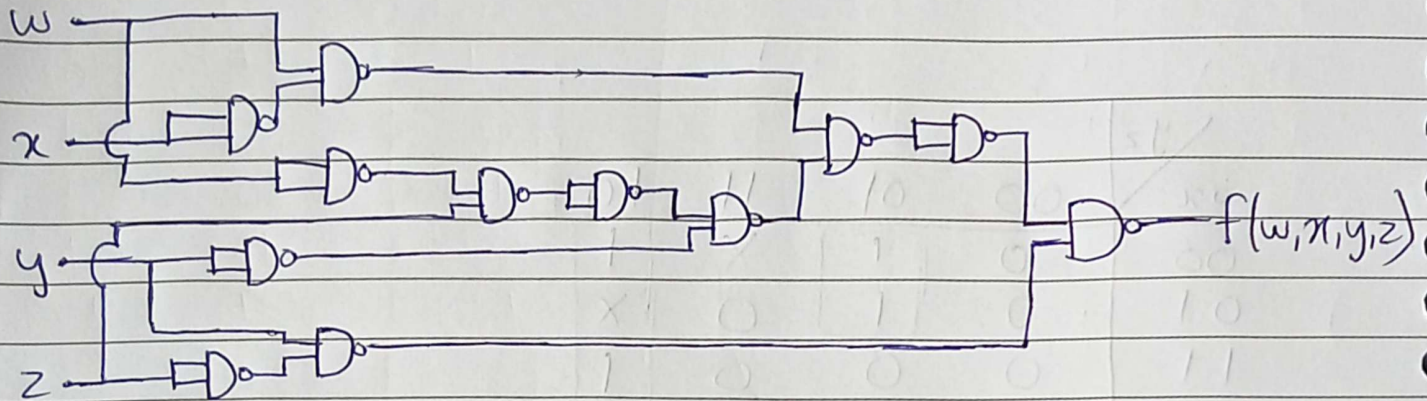
Terms = 6

Literals = 24

Simplified $\bar{x}w + y\bar{z} + \bar{w}\bar{y}z \rightarrow 7 \text{ literals.}$

d) NAND implementation.

$$w\bar{x} + y\bar{z} + \bar{w}y\bar{z}$$



• Completes part 8(a)

adding B, C in $(\bar{A} + \bar{D})$:-

$$= (\bar{A} + \bar{D} + B + C) (\bar{A} + \bar{D} + \bar{B} + C) (\bar{A} + \bar{D} + B + \bar{C}) (\bar{A} + \bar{D} + \bar{B} + \bar{C})$$

adding A, D in $(C + B)$:-

$$= (A + D + C + B) (\bar{A} + \bar{D} + C + B) (A + \bar{D} + C + B) (\bar{A} + \bar{D} + C + B)$$

$$\text{all terms} = (A + B + \bar{C} + D) (A + B + C' + D') (A + B' + C + D) (A + \bar{B} + C + D) \\ (A + \bar{B} + C + D) (A + \bar{B} + C' + D') (A' + B + C' + D) (A' + B' + C + D) \\ (A' + B' + C' + D)$$